lecture11 Practice Questions and Solutions

Question 1 (q1_sol.py)

Make a fortune cookie program that, when run, randomly prints one of these phrases:

- Whatever you do, always give 100% — unless you're donating blood.
- Be a cupcake in a world full of muffins.
- We become what we think about: so don't think too much about hot dogs.
- You can’t have everything. Where would you put it?

Sample run 1

```
Be a cupcake in a world full of muffins.
```

Sample run 2

```
You can't have everything. Where would you put it?
```

Sample run 3

```
Be a cupcake in a world full of muffins.
```

Since the program chooses the quote randomly, there will be repeats across different runs.

Sample Solution

```
import random

n = random.randint(1, 4)
if n == 1:
    print("Whatever you do, always give 100%—unless you're donating blood.")
elif n == 2:
    print("Be a cupcake in a world full of muffins.")
elif n == 3:
    print("We become what we think about: so don't think too much about hot dogs.")
elif n == 4:
    print("You can't have everything. Where would you put it?")
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter the side-length of a perfectly square field, and then prints the area and the perimeter of the field.

However, you must do it by creating and using two functions: one that calculates the area of the square (given the side length), and one that calculates the perimeter of the square (given the side length). Each of these functions should return (not print!) their result.

Use these functions in a sensible way in your program.

```
How long is the field? 6

The area is 36.
The perimeter is 24.
```

Sample Solution

```
def square_area(size):
    return size * size

def square_perimeter(size):
    return 4 * size

size = float(input("How long is the field? "))
area = square_area(size)
perimeter = square_perimeter(size)
print(f"The area is {area}.")
print(f"The perimeter is {perimeter}.")
```

Question 3 (q3_sol.py)

Write a program that asks the user to enter an integer, and then prints the times-table for that integer from 1 to 12 as shown in the example.

Sample Run 1

```
Enter an integer: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
5 x 11 = 55
5 x 12 = 60
```

Sample Run 2

```
Enter an integer: 43
43 x 1 = 43
43 x 2 = 86
43 x 3 = 129
43 x 4 = 172
43 x 5 = 215
43 x 6 = 258
43 x 7 = 301
43 x 8 = 344
43 x 9 = 387
43 x 10 = 430
43 x 11 = 473
43 x 12 = 516
```

Sample Solution

```
n = int(input("Enter an integer: "))
for i in range(1, 13):
    print(f"{n} x {i} = {n * i}")
```

Question 4 (q4_sol.py)

During the summer, a neighborhood has these rules for lawn watering:

- houses with even street numbers may water their lawn only on even-numbered days
- houses with odd street numbers may water their lawn only on odd-numbered days

Write a program that asks the user to enter a street number (like 503 or 1944) and a day number (Monday is 1, Tuesday is 2, ..., Sunday is 7), and then prints whether or not that house may water their lawn that day.

As an extra challenge, write and use a function called `can_water(street_number, day)` that returns `True` if the house may water their lawn that day, and `False` otherwise.

Sample run 1

```
Enter a street number: 503
Enter a day number: 1
You can water your lawn today.
```

Sample run 2

```
Enter a street number: 1944
Enter a day number: 2
You can water your lawn today.
```

Sample run 3

```
Enter a street number: 1944
Enter a day number: 3
You cannot water your lawn today.
```

Sample Solution

```
def can_water(street_number, day):
    if street_number % 2 == 0:
        return day % 2 == 0
    else:
        return day % 2 != 0

street_number = int(input("Enter a street number: "))
day = int(input("Enter a day number: "))
if can_water(street_number, day):
    print("You can water your lawn today.")
else:
    print("You cannot water your lawn today.")
```